National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	**Computer Networks**	**Course Code:**	**CS 3001**
	Program:	**BS (Computer Science)**	**Semester:**	**Spring 2026**
	Duration:	**15 minutes**	**Total Marks:**	**15**
	Paper Date:		**Section**	**6B1**
	Exam Type:	**Quiz 2 - Chapter 2**	**Page(s):**	**2**

Student Name **Roll No.** **Section:**

Q1. Carefully read the Question before attempting: **[15 marks]**

An automated rover on Mars needs to download a firmware update from a server on Earth. The update consists of one base HTML index file (size L_b = 2 KB) and 8 referenced binary objects (each size L_o = 100 KB). The link characteristics are:

- Distance Earth-Mars: d = 2.25×10^{11} meters.
- Transmission Rate: R = 1 Mbps.

The rover opens parallel TCP connections but is limited to a maximum of K = 3 parallel connections due to memory constraints. Calculate the total time elapsed from the initial connection request until the last object is fully received. Assume:

- Short control packets (SYN, ACK, HTTP GET) have negligible transmission time compared to propagation delay.
- We are using Non-Persistent HTTP over the parallel connections. No DNS lookup is required (IP is hardcoded). Speed of light c = 3×10^8 m/s.